Faculty of Computer & Information Sciences
Ain Shams University
Course Code: SCO 421
Course Name: Computer Vision
Exam: (MidTerm) 16/11/2022
Year: (1st term) 4th undergraduate (SC-CSys)

Instructors: Dr. Dina Khattab
Offering Dept: Scientific Computing
Academic year: 2022-2023
Duration: 45 Minutes
Total Grade: 15 Marks

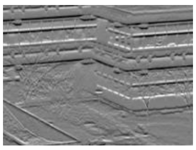
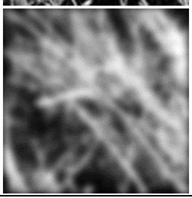
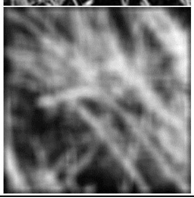
....			الاسم:
....	رقم الجلوس:	....	القسم:

A) Given the following table showing sets of input and output images: <u>(5 Marks)</u>

i) Define the appropriate image operation used to produce each corresponding output.

ii) Select the related image filter to each operation from the following set of filters

$\begin{matrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{matrix}$	$\begin{matrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{matrix}$	$\begin{matrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{matrix}$	$\begin{matrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{matrix}$
(a)	**(b)**	**(c)**	**(d)**
$\begin{matrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{matrix}$	$\begin{matrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{matrix}$	$\begin{matrix} -1/9 & -1/9 & -1/9 \\ -1/9 & 17/9 & -1/9 \\ -1/9 & -1/9 & -1/9 \end{matrix}$	$\begin{matrix} 1/16 & 1/8 & 1/16 \\ 1/8 & 1/4 & 1/8 \\ 1/16 & 1/8 & 1/16 \end{matrix}$
(e)	**(f)**	**(g)**	**(h)**

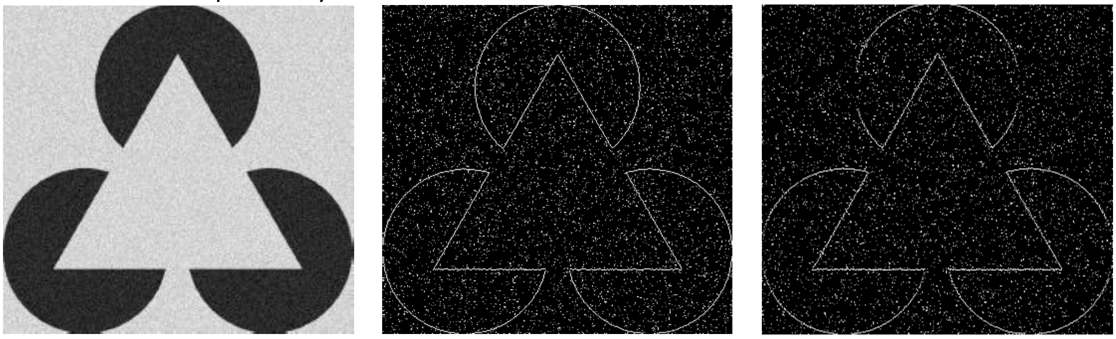
Image set	**1**	**2**	**3**	**4**	**5**
Input					
Output					
Image operation					
Filter					

B) Consider the task of edge detection in linear images: <u>(3 Marks)</u>

i) Compare between **Canny** edge detector and the **Laplacian-of-Gaussian (LoG)** edge detector in terms of the order of the derivatives that it computes and the number of convolution operations it requires.

	Canny	**Laplacian-of-Gaussian (LoG)**
Order of the derivative		
Number of convolution operations		

ii) Which of the following images (a) or (b) is more likely to be produced by Canny edge detector? Explain why?



Original Image **(a)** **(b)**

C) In the **Harris** feature detector, given the following second-moment matrix $M = \begin{bmatrix} 403 & 385 \\ 385 & 381 \end{bmatrix}$, compute the Harris response function R with $\alpha = 0.04$. Do you think that this value indicates the existence of a corner, edge or a flat area? (2 Marks).

D) Many Computer Vision algorithms such as **SIFT** (Scale-Invariant Feature Transform) seek to detect and analyze features at multiple scales of analysis. (5 Marks).

i) Explain briefly what is "scale-space".

ii) For what operation is the Difference-of-Gaussian (DoG) used in different scale levels? Show its advantage over Laplacian-of-Gaussian (LoG).

iii) A SIFT descriptor for a 16×16 image window W is a vector of 128 numbers, which can be thought of as being grouped in 16 consecutive groups of 8 numbers each. What does each group of 8 numbers represent?

With Best Regards,

Dr. Dina Khattab